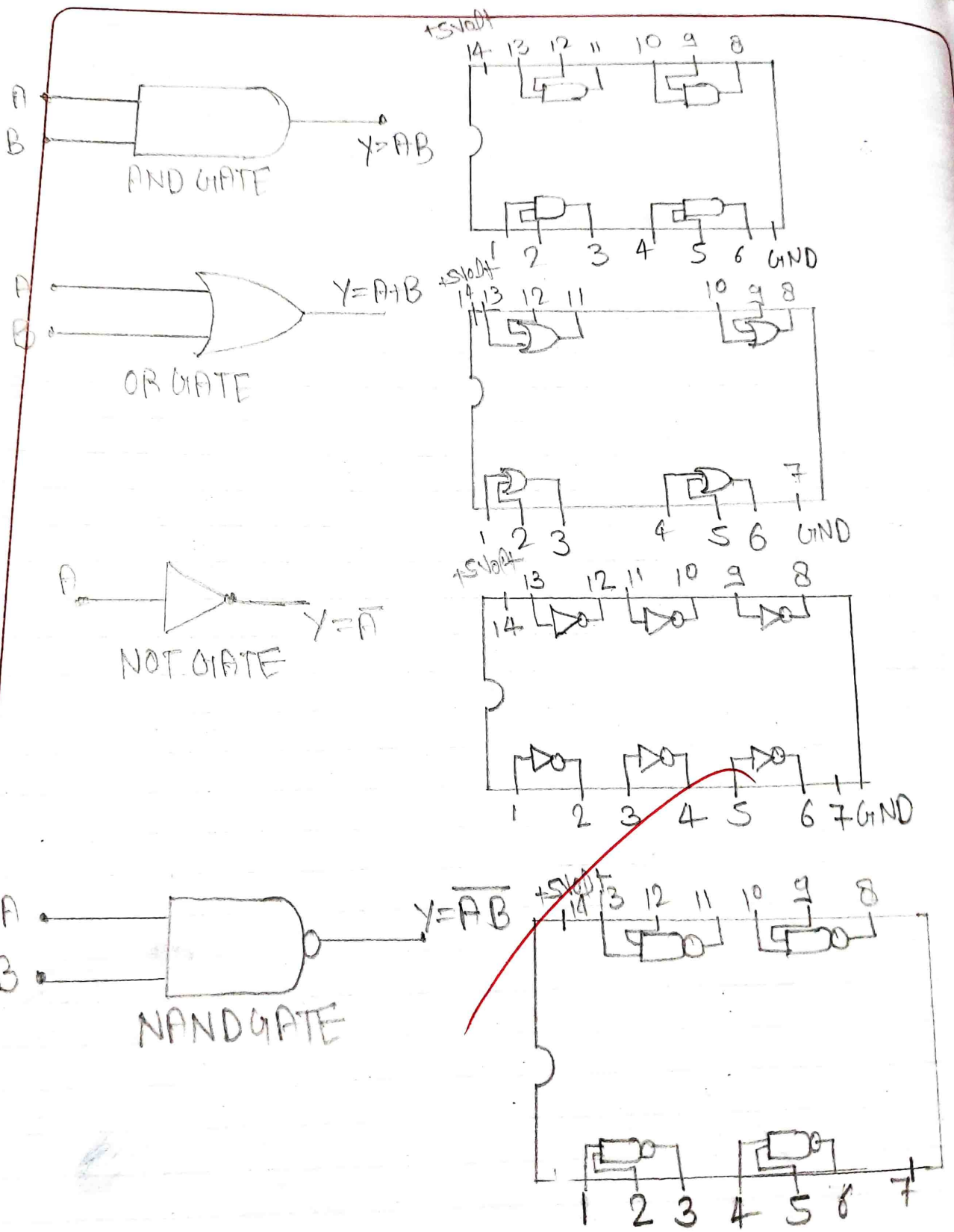
# INDEX

Name : Kanishk Patel Class & Section : BTech CSE (D2)

Subject : Computer Organization LAB Roll No. 26

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| 8 | Design of counter (Mod-10) | 23 | 17-11-22 | | |
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A
B
AND GATE
Y=AB
+5VDC
14 13 12 11 10 9 8
1 2 3 4 5 6 GND
A
B
OR GATE
Y=A+B
+5VDC
14 13 12 11 10 9 8
1 2 3 4 5 6 7 GND
A
NOT GATE
Y=Ā
+5VDC
14 13 12 11 10 9 8
1 2 3 4 5 6 7 GND
A
B
NAND GATE
Y=$\overline{AB}$
+5VDC
14 13 12 11 10 9 8
1 2 3 4 5 6 7

Name of Experiment..... Date.....

Experiment No. 1 Experiment Result..... Page No. 1

Objective – Bread board implementation of different AND, OR, NOT, NAND, NOR and XOR.

Apparatus required - Bread board wires Logic trainers and ICs.

Theory -

1. AND gate (IC-7408) -

The AND gate is a logic circuit that has two or more inputs and one output. If any of the input is 0 or (low) then the output will be 0 and if both the input are low or high the output will be corresponding low or high.

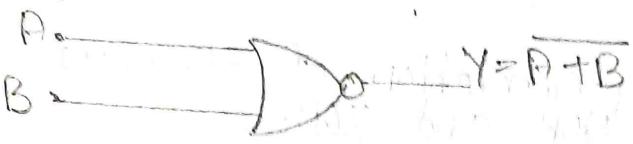
2. OR Gate (IC-7432) -

The OR Gate has been two or more inputs but only one output. If any input signal is high or 1 the output is one and if both the input is 0 then output will be 0. The OR gate can have as many inputs as desired.

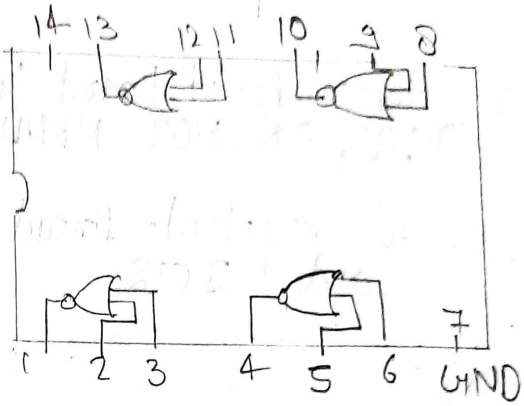
3. NOT Gate (IC-7404) -

The NOT gate is the simplest form of digital logic circuit. The other name of this gate is inverter gate. The output state is always opposite to the input state. If input is 0 then the output is 1 and vice versa.

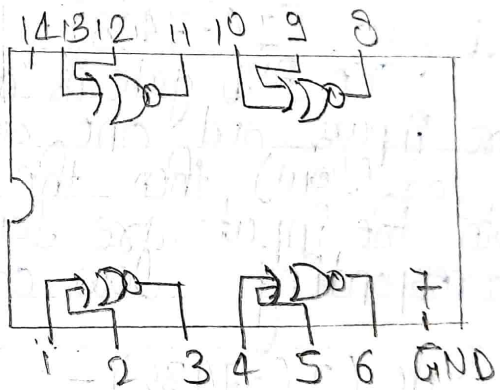
Teacher's Signature :


A
B
Y = $\overline{A+B}$

NOR GATE


14 13 12 11 10 9 8
1 2 3 4 5 6 7
GND


A
B
Y = A⊕B


14 13 12 11 10 9 8
1 2 3 4 5 6 7
GND

4. NAND Gate (IC 7400) - The NAND Gate is an improved logic element used for implementing decision making logic functions. The NAND Gate is in fact a NOT-AND Gate.

5. NOR Gate (IC - 7402) - The NOR gate is an improved logic element used for implementing decision making logic functions. The nor Gate is in fact a NOT OR gate.

6. XOR Gate (IC - 7486) -
The XOR gate work as logical "either/or". The output is high if either input is high and output is low when either both input are high or both are low

Observation Table -

AND gate truth table

| Input | | Output |
|---|---|---|
| A | B | Y = A.B |
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

Teacher's Signature :

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No 3

OR gate truth table-

| Input | | Output |
|---|---|---|
| A | B | Y=A+B |
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

NOT gate truth table.

| Input | Output |
|---|---|
| A | ~~Output~~ |
| 0 | 1 |
| 1 | 0 |

NAND gate truth table.

| Input | | | Output |
|---|---|---|---|
| A | B | AB | Y= AB |
| 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

Teacher's Signature :

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No 41

NOR gate truth table -

| Input | | | Output |
|---|---|---|---|
| A | B | A+B | $Y = \overline{A+B}$ |
| 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 |

XOR gate truth table

| Input | | Output |
|---|---|---|
| A | B | $Y = A \oplus B$ |
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

Result - All the truth tables of all the basic gates are verified.

Precautions -

1. All connections should be tight.
2. All instruments should be of proper range.

Teacher's Signature :

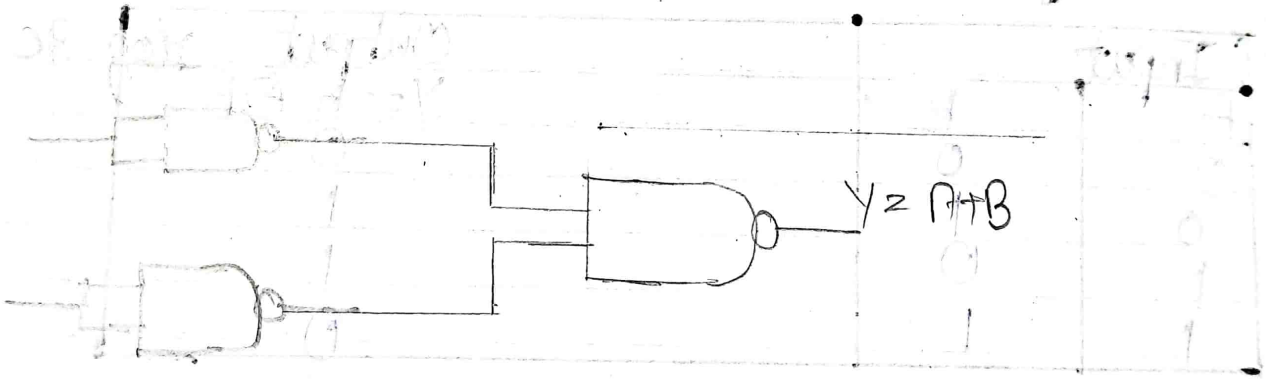
1. Implementation using NAND gate

(a) NOT gate

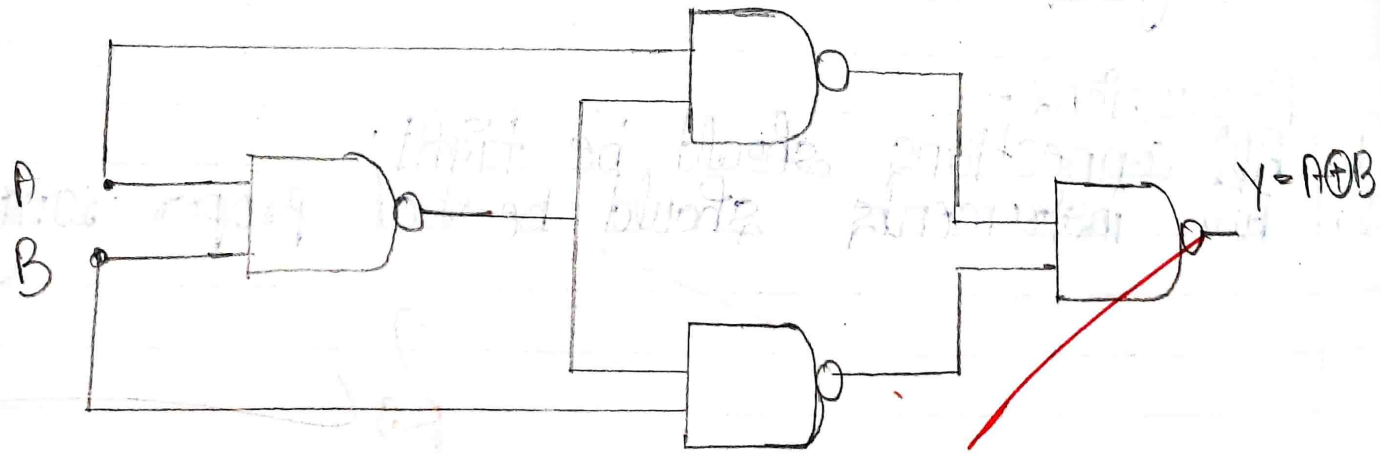



(b) AND gate




(d) : XOR gate



Name of Experiment..... Date.....

Experiment No. 2 Experiment Result..... Page No. 5

# Experiment -2

Object- To realize all logic gates using NAND and NOR gates.

Apparatus required- IC 74002, IC 7400, Digital Trainer kit, Bread board.

Observation Table-

1. Implementation using NAND gate-

a) NOT gate-

| A | Y=A' |
|---|---|
| 0 | 1 |
| 1 | 0 |

(b) AND gate

| A | B | Y=A·B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

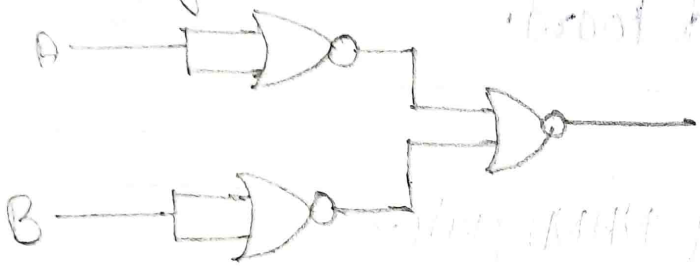
Teacher's Signature :

2. Implementations using NOR gate

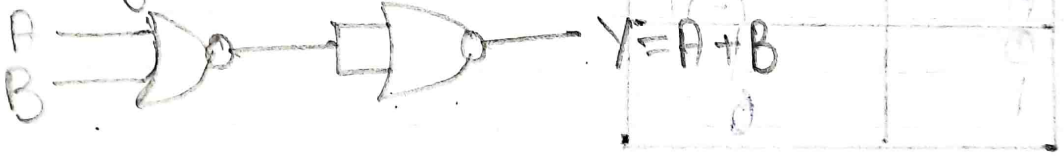
(a) NOT gate



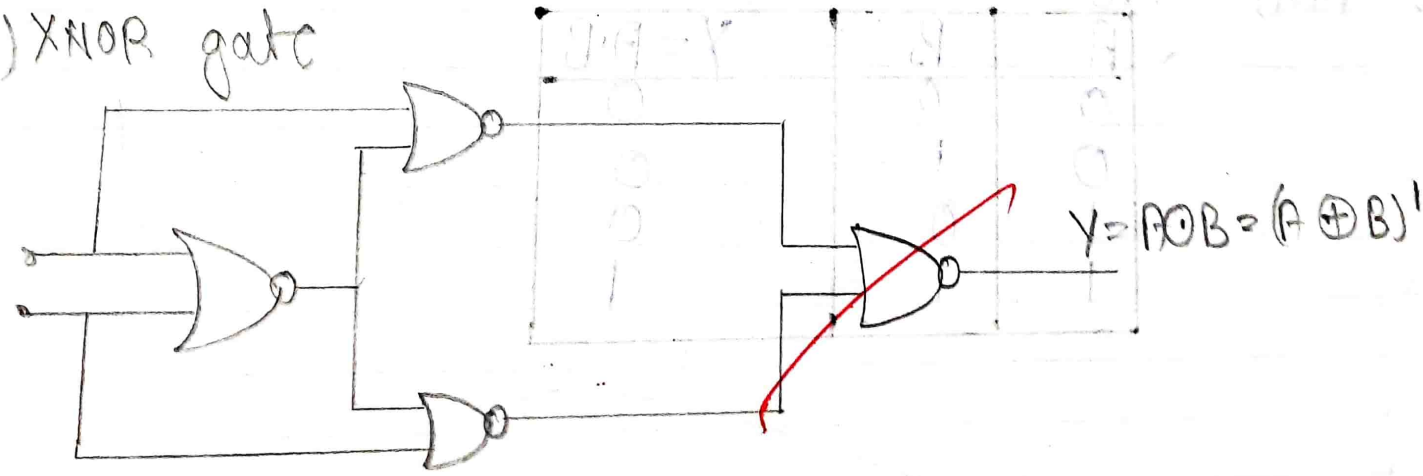

(b) AND gate




(c) OR gate




(d) XNOR gate



c) OR gate-

| A | B | Y=A+B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

d) XOR gate-

| A | B | Y= A⊕B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

2. Implementation using NOR gate-

a) NOT gate-

| A | Y=$\bar{A}$ |
|---|---|
| 0 | 1 |
| 1 | 0 |

(b) AND gate-

| A | B | Y=A·B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

Teacher's Signature : _____

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 7

c) OR gate

| A | B | Y=A+B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

d) XNOR gate-

| A | B | Y = A⊙B |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

Precautions -

1. Connections must be right on the bread board
2. Identify the pins IC correctly
3. Take care while removing and inserting the IC on bread board.

Result- Realized all basic logic gates using NAND or NOR gates.

Teacher's Signature :

Name of Experiment..... Date.....

Experiment No. 3 Experiment Result..... Page No 8

# Experiment no-3

Objective- Implementation of Half adder and Full adder using logic gates.

Apparatus required-

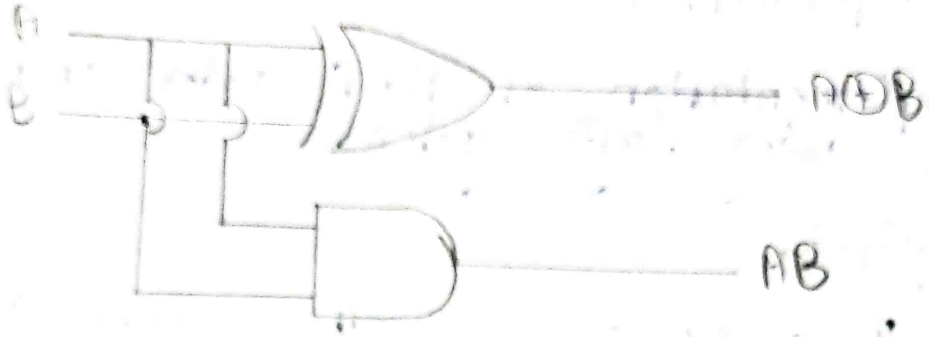
| S.No | Apparatus | Specification | QTY |
|---|---|---|---|
| (i) | IC-7408 | AND GATE | 1 |
| (ii) | IC-7432 | OR GATE | 1 |
| (iii) | IC-7486 | XOR GATES | 1 |
| (iv) | Bread board | – | 1 |
| (v) | Jumper wires | – | As per |
| (vi) | LED | – | 1 |
| (vii) | Power Supply | +5V | – |

Theory-

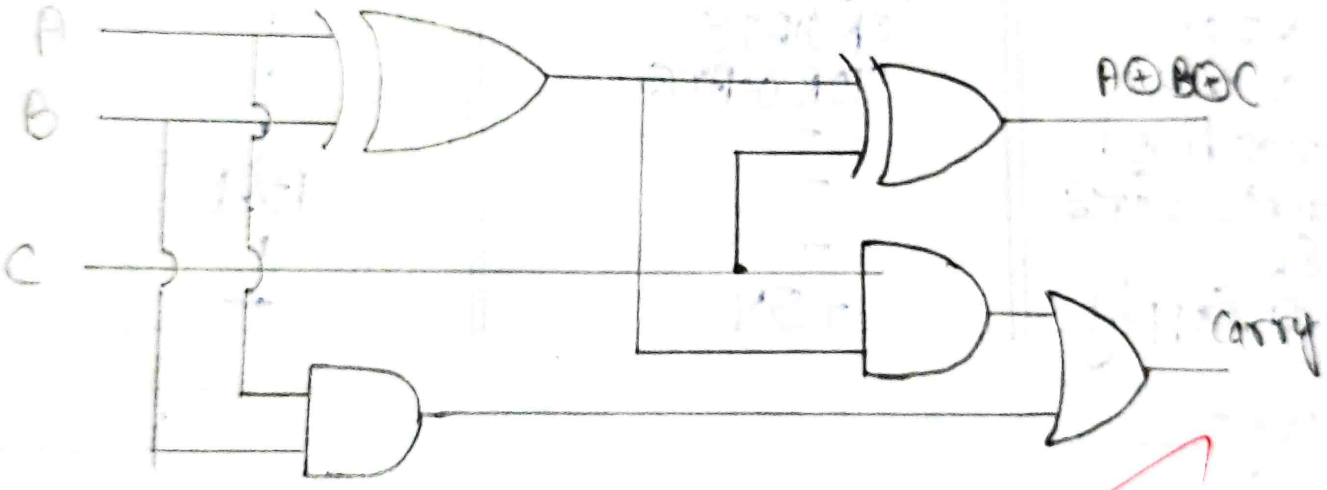
(i) Half Adder- The most basic digital arithmetic circuit is the addition of two binary digits A combinational circuit that performs the arithmetic addition of two bits is called a Half Adder. The input variables of a half Adder are called the augend and addend bits. The output variables the sum and carry.

Teacher's Signature :

## Circuit diagram of Half Adder-



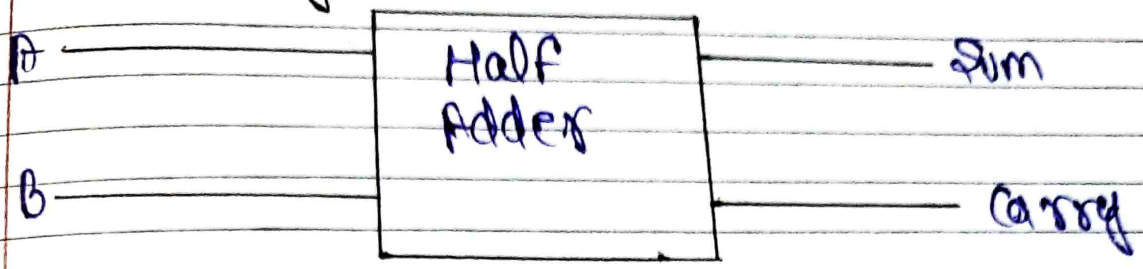

## Circuit Diagram of full Adder-



Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 9

Block diagram -




Truth table -

| A | B | Sum | Carry |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

Equation:

$Sum = \bar{A}B + A\bar{B} = A \oplus B$

$Carry = AB$

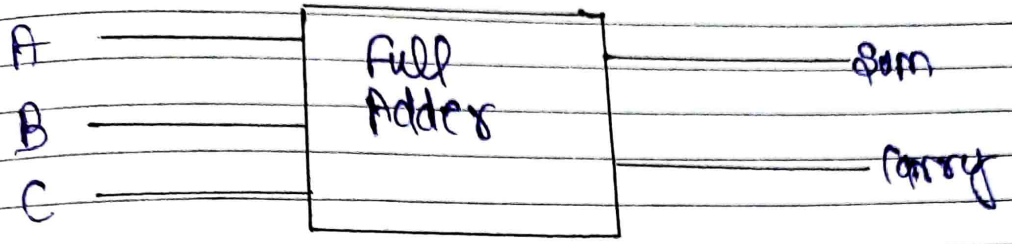
2. Full Adder - A full adder is a combinational circuit that forms the arithmetic sum of the three input bits. It consists of three inputs and two outputs. So that two half adder is required to implement a full adder.

Teacher's Signature :

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No 13

Block diagram:-




Truth table -

| A | B | C | Sum | carry |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

Equation -

$Sum = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$

$= \bar{A}(\bar{B}C + B\bar{C}) + A(\bar{B}\bar{C} + BC)$

$= \bar{A}(B \oplus C) + A(\overline{B \oplus C})$

$= A \oplus B + C$

$Carry = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$

$= C(\bar{A}B + A\bar{B}) + AB(\bar{C} + C)$

$= C(A \oplus B) + AB$

Teacher's Signature :

Name of Experiment..... Date.....

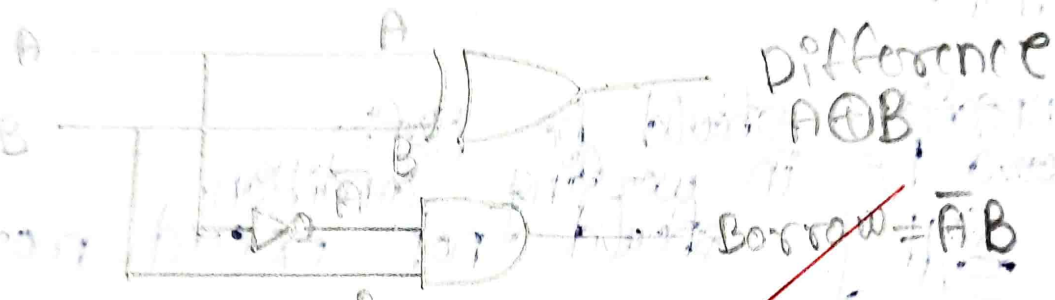
Experiment No..... Experiment Result..... Page No. 41

Result - Both Half adder and full adder are implemented using different logic gates.

Precaution -

(i) All connection should be tight.
(ii) IC should be in working condition.
(iii) Power supply should not exceed more than 5V.

Teacher's Signature :




Half Subtractor

Name of Experiment..... Date.....

Experiment No. 4 Experiment Result..... Page No. 12

Objective- To implement half subtractor and full subtractor

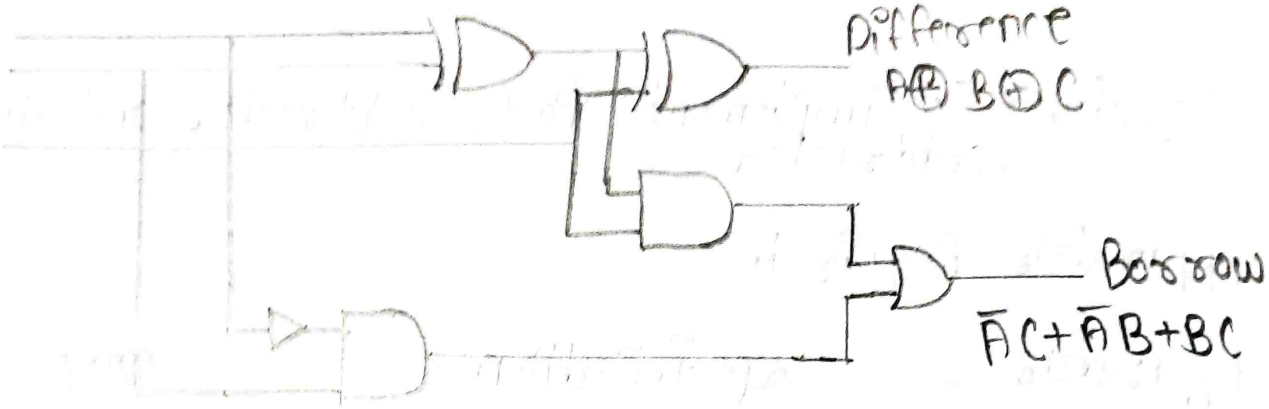
Apparatus Required.

| Apparatus | specification | QTY |
|---|---|---|
| (i) AND Gate | IC 7408 | 1 |
| (ii) X-OR Gate | IC 7486 | 1 |
| (iii) NOT Gate | IC 7404 | 1 |
| (iv) OR Gate | IC 7432 | 1 |
| (v) Powerful supply +5V | | 1 |
| (vi) Bread Board | | 1 |
| (vii) LED | | ADEQUATE |
| (viii) Connecting wires | | as required |

Theory -

Half Subtractor- The half subtractor constructed using XOR and AND Gate. The half subtractor has two input and two outputs. The output are difference and borrow. The difference can be applied using XOR Gate borrow output can be implemented using an AND Gate and an inverter.

Teacher's Signature :




Full Subtractor

Truth Table

| A | B | Borrow | Difference |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 0 |

Full subtractor - The full subtractor is a combination of XOR, AND, OR, NOT gates. In a full subtractor the logic circuit should have three inputs and two outputs. The two half subtractor put together gives a full subtractor. The first half subtractor will be C and AB. The output will be difference output of full subtractor. The expression AB assembles the borrow output of the half subtractor and the second term is the inverted difference output of first X-OR.

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No 14

Truth table.

| A | B | C | Borrow | Difference |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |

Result - The half subtractor and full subtractor circuits are implemented and verified

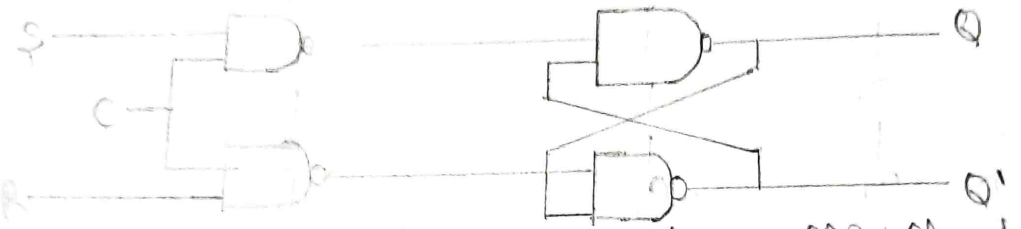
Precaution -

(i) Switch off the power supply while making connections.

(ii) All the connections should be made properly

(iii) Get your circuit checked either by faculty or by Lab assistance before you switch on the power supply.

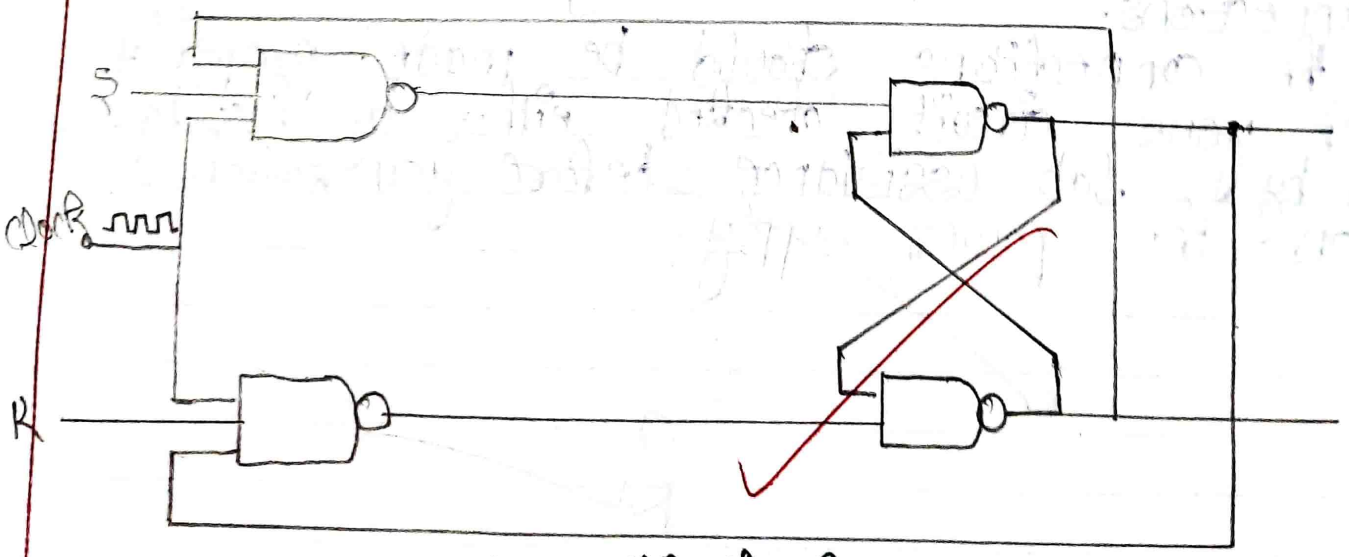
Teacher's Signature :




(Logic diagram of SR-flip flop)

Truth table

| C | S | R | Q |
|---|---|---|---|
| 0 | X | X | No change |
| 1 | 0 | 0 | No change |
| 1 | 0 | 1 | Q = 0; Reset state |
| 1 | 1 | 0 | Q = 1; Set state |
| 1 | 1 | 1 | Indeterminate |




(JK flip flop)

Name of Experiment..... Date.....

Experiment No. 5 Experiment Result..... Page No. 15

# Experiment - 5

Objective - Bread board implementation of flip flop (SR, JK, DT)

Apparatus required - NAND gate, power supply, Bread board, LED, wires

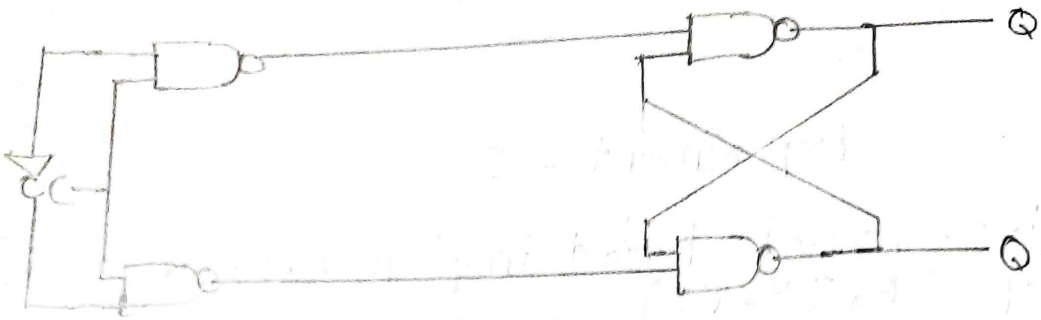
Theory - Flip flop are basic fundamental building block of digital system. Flip flop is binary storage device that can store only 1 bit of data. It has property to remain same in one state until it is directed by input signal to switch over the state.

There are 4 types of flip flops

SR (Set Reset)
D (Data or Delay)
T (Toggle)
JK.

SR flip flop - SR flip flop consists of two cross coupled NAND gate. A clocked SR flip flops has an additional clock input so that S and R input are active only when clock is high when clock goes low state of flip flops can not change until clock goes to high.

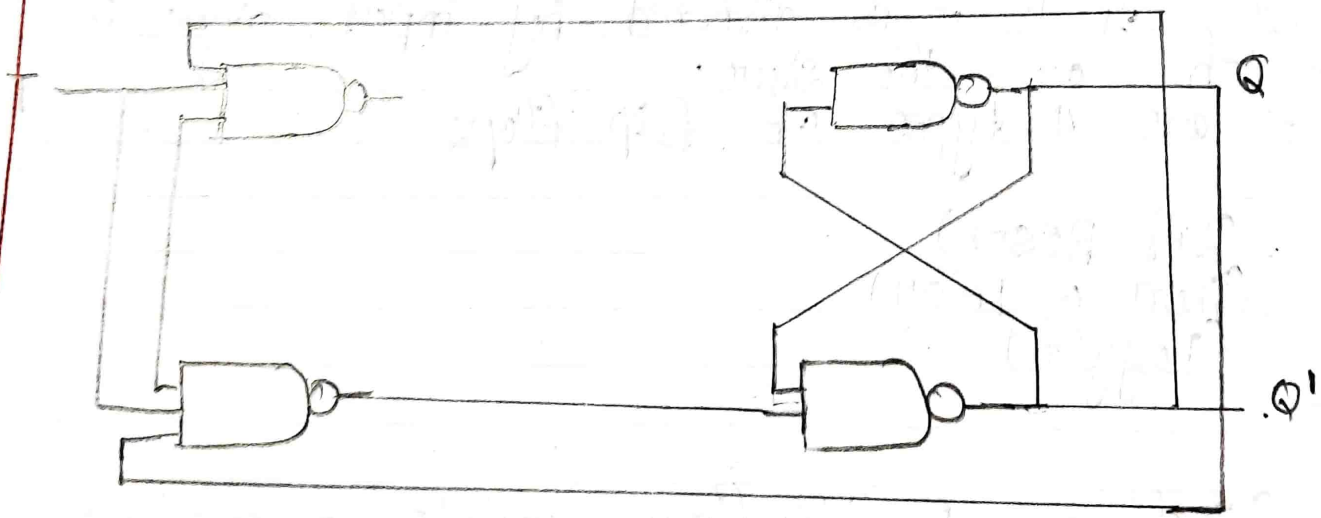
Teacher's Signature :




(D- Flip Flop)

Truth table -

| C | D | Q |
|---|---|---|
| 0 | X | X (No change) |
| 1 | 0 | Q = 0, Reset |
| 1 | 1 | Q = 1 Set |




Truth table -

| T | Qn | Qn+1 |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 16

JK flip-flop - JK flip flop is similar to SR flip flop has two basic problem then to overcome these problem, JK flip flop was develop.

In J.K flip flop two input NAND gate are required with three input NAND gate. Third input of each gate is connected to the output of Q and Q'

Tooth table for JK flip flops -

| J | K | Clock | Q |
|---|---|---|---|
| 0 | 0 | ↑ | $Q_0$ (no change) |
| 0 | 1 | ↑ | 0 |
| 1 | 0 | ↑ | 1 |
| 1 | 1 | ↑ | $\bar{Q}_0$ (toggle) |

D flip flop - D flip flop is most important flip flop from other clocked types. It ensure that same time both input never equals to 1. It is designed using gate. SR flip flop with an inverter connected between input allowing for a signal input. D is refused as Data, when data input set 1 flip flop would be set, when D set to 0 flip flop would change.

Teacher's Signature : _____

Name of Experiment..... Date.....

Experiment No..... Experiment Result.....

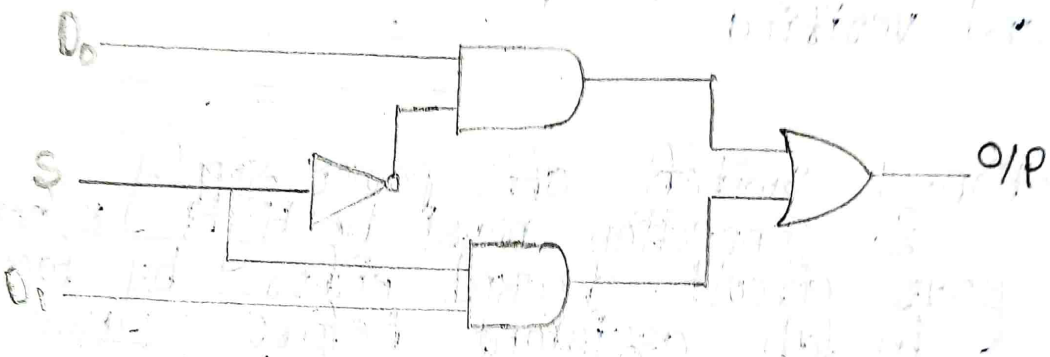


T flip flop - T stands for toggle. T flip flop has only one input constructed by connecting input of JK flip flop.

Result - Various flip flop has been implemented and verified

Precaution - 1. Switch off power supply
2. Connection must be tight & neat
3. Get your circuit checked either by faculty or by lab assistant before you switch on the power supply.

Teacher's Signature : _____




Logic circuit of two to One Line Mux

Truth table

| A | B | C | F |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

Name of Experiment..... Date.....

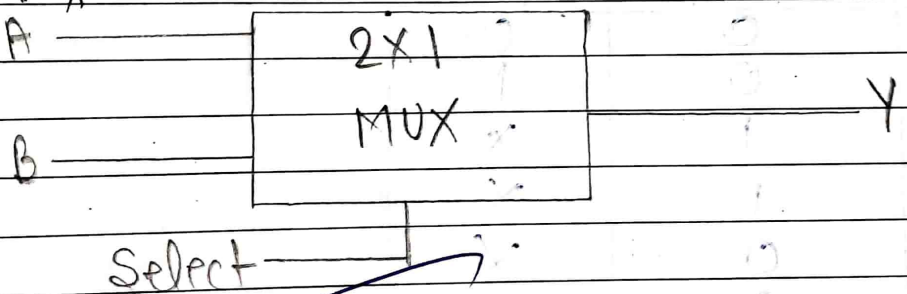
Experiment No. 6 Experiment Result..... Page No. 18

Objective - Design 2X1 and 4X1 multiplexer using logic gates.

Theory -

* Multiplexer - A multiplexer is a network that has many inputs and one output and the value of the output will be the value of one of inputs which will be decided by some select lines. The simplest type of multiplexer is the two line to one line data multiplexer.

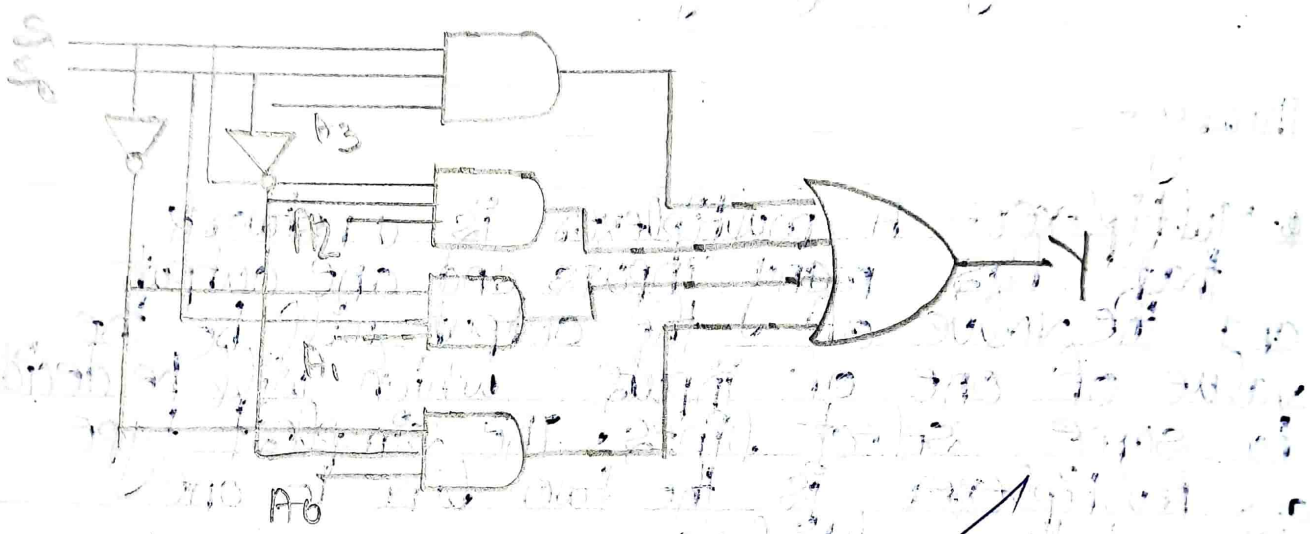
2X1 MUX -




Expression - $F(A,B,C) = \bar{A}C + \bar{B}C + AB\bar{C}$

Teacher's Signature : _____

4 to 1 multiplexer circuit diagram-




Truth table-

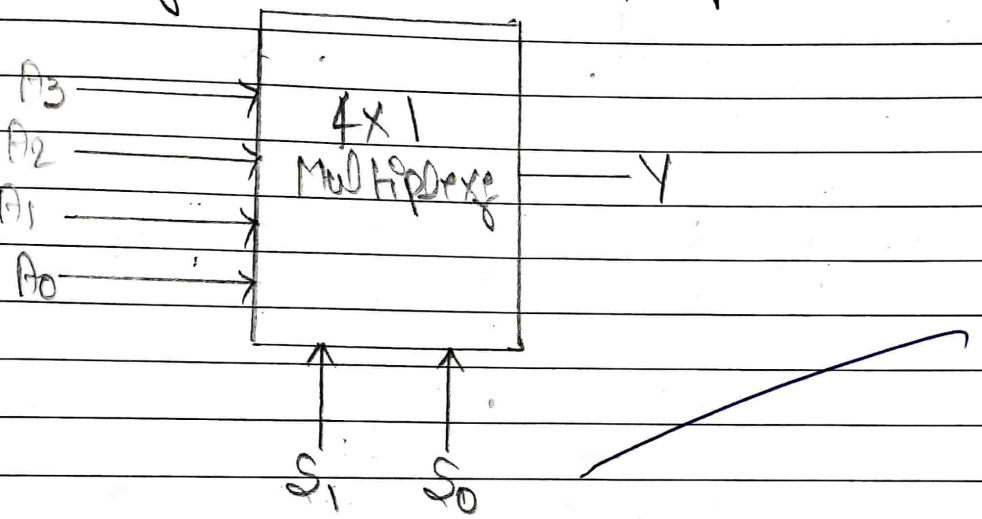
| $S_0$ | $S_1$ | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 1 | x |
| 0 | 1 | x |
| 1 | 0 | x |
| 1 | 0 | x |
| 1 | 1 | x |
| 1 | 1 | x |

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 19

4 X 1 Multiplexer - A 4 to 1 MUX contain four input lines and these are $D_0$ $D_1$ $D_2$ and $D_3$ two selected lines $S_0$ and $S_1$ and one output Y-line.

Block diagram of 4 to 1 Multiplexer -




Expression -

$$Y = S_1\bar{S}_0A_0 + \bar{S}_1S_0A_1 + S_1\bar{S}_0A_2 + S_1S_0A_3$$

Result - 2 to 1 and 4 to 1 multiplexer implemented successfully.

Teacher's Signature : _____

Name of Experiment..... Date 10-11-22

Experiment No. 7 Experiment Result..... Page No. 20

# Experiment No-7

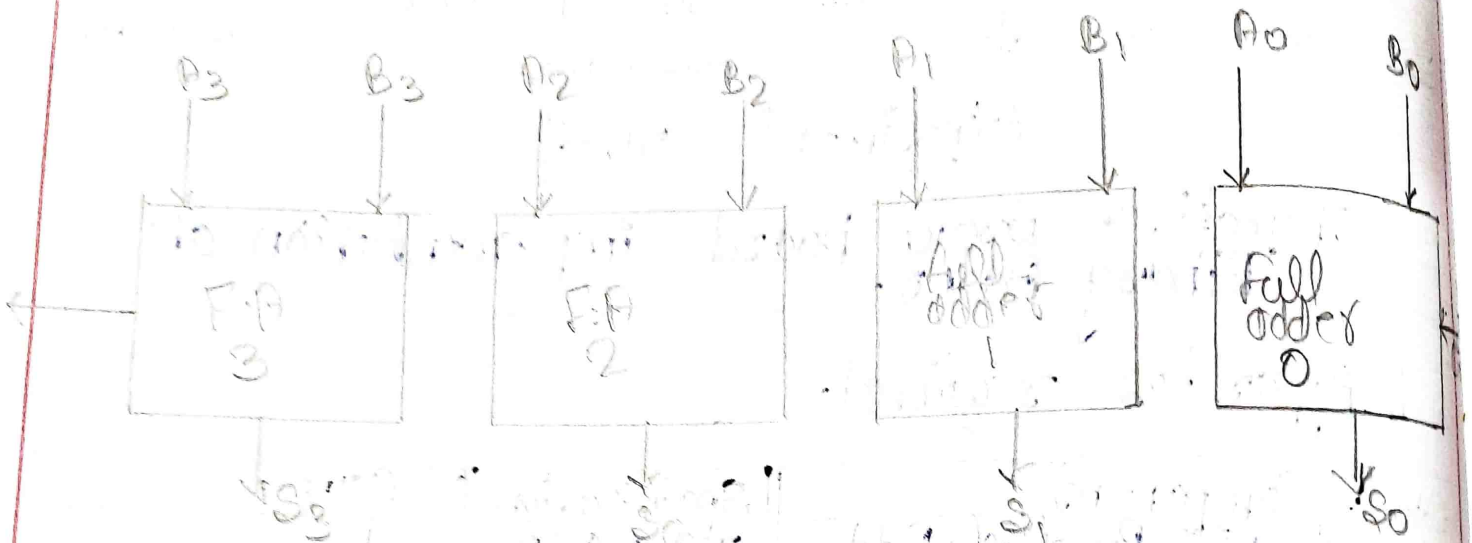
Objective- Bread board implementation of Binary Adder.

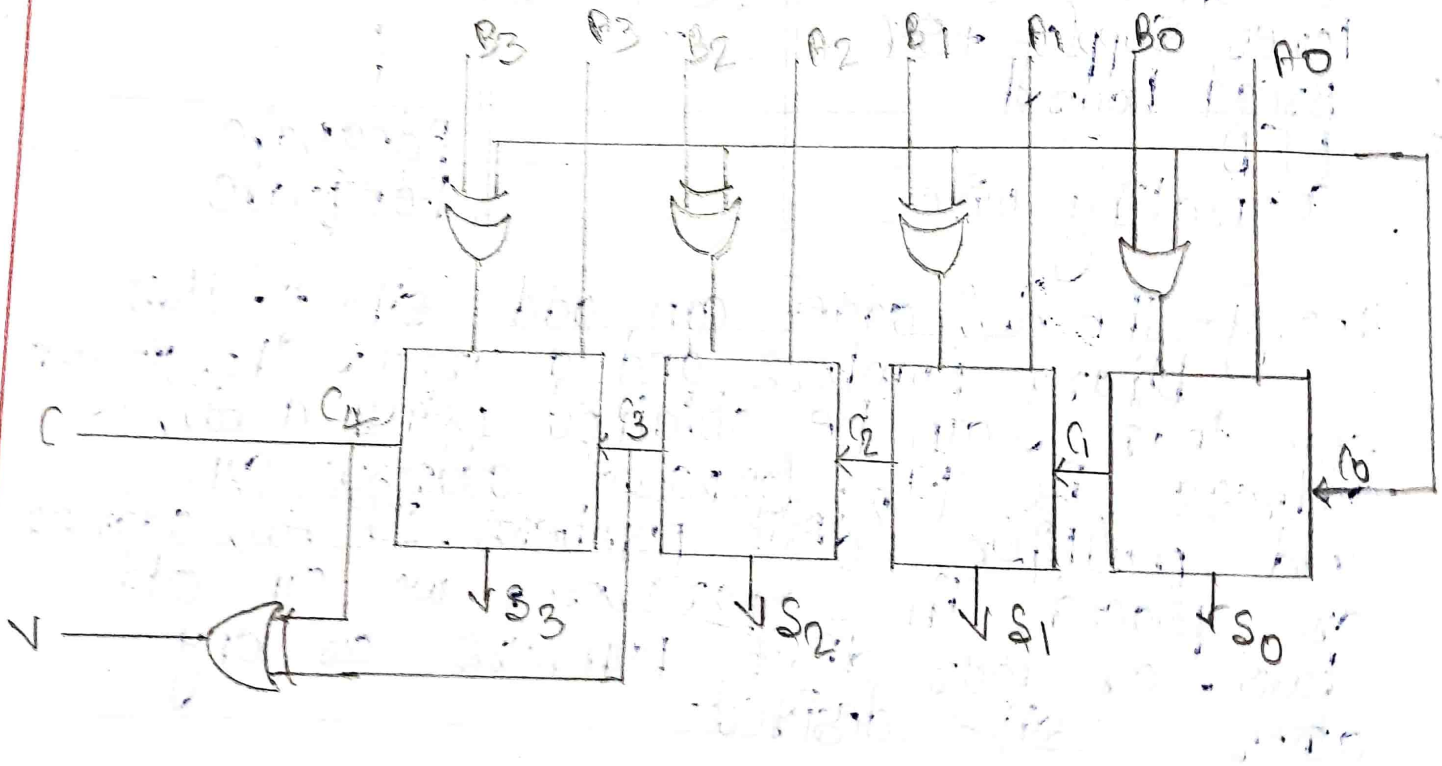
Apparatus required.

| S.No | Component | Specification | Qty |
|---|---|---|---|
| 1 | 4-BIT Parallel Adder | IC 7483 | 1 |
| 2 | Ex-OR | IC 7486 | 1 |
| 3 | Power Supply +5V | — | 1 |
| 4 | Bread board | — | 1 |
| 5 | LED | | Adequate |
| 6 | Connecting wire | | Adequate |

Theory- The full adder can add single-digit binary numbers and a carry. The largest sum that can be obtained using a full address is 1 1/2. Parallel address can add multiple digit numbers. If full address are placed in parallel, we can add two- or four digit numbers or any other size desired.

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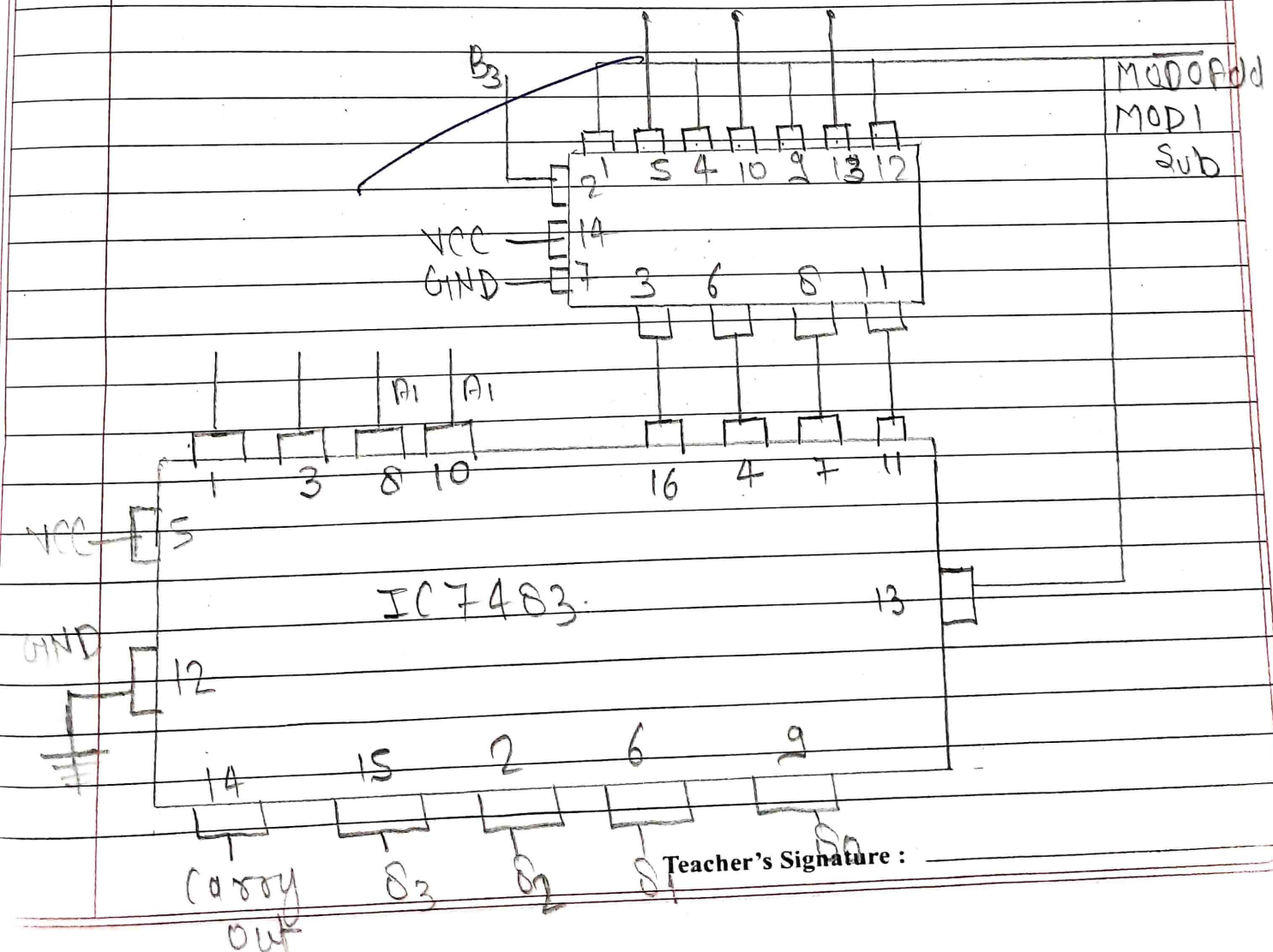

A3
B3
A2
B2
A1
B1
A0
B0
F.A 3
F.A 2
Full adder 1
Full adder 0
S3
S2
S1
S0


B3
A3
B2
A2
B1
A1
B0
A0
C
C4
3
C2
C1
C0
V
S3
S2
S1
S0

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 21

Subtraction is performed adding 2's complement of augend and addend by simply taking 1's complement of augend and taking input carry 1. Common circuit can be used to perform addition as well as subtraction one input of each Ex-OR gate and input carry to a common terming known as mode input. This mode input decides whether circuit will perform addition or subtraction. If Mode = 0, it perform addition if Mode = 1 it perform subtraction.




Teacher's Signature : _____

Procedure-

1. Connections are given as per circuit diagram
2. Logical inputs are given as per circuit diagram
3. Observe the output & verify the truth table

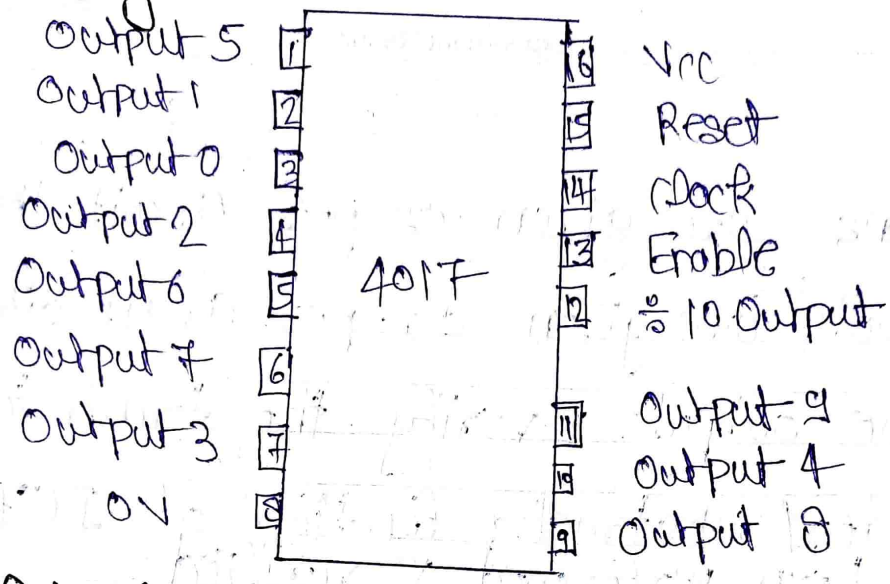
Result- Adder/Substractor function of IC7483 has been performed & verified.

Precaution-

1. Switch off the power supply while making connections.
2. All the connections should be made properly as per the circuit diagram.
3. Get your circuit checked either by faculty or by lab assistant before you switch on the power supply.

Teacher's Signature :

Pin diagram-




Truth table-

| Count | $Q_9$ | $Q_6$ | $Q_7$ | $Q_6$ | $Q_5$ | $Q_4$ | $Q_3$ | $Q_2$ | $Q_1$ | $Q_0$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 3 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 4 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 5 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 6 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 7 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 9 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Name of Experiment..... Date 17-11-22

Experiment No..... Experiment Result..... Page No. 23

# Experiment no- 8

**Objective-** Design of counter (Mod - 10)

**Apparatus Required -**

| S.No | Component | Specification | QTY |
|---|---|---|---|
| 1 | Counter | IC 4017 | 1 |
| 2 | Power supply +5V | — | 1 |
| 3 | Bread board | — | 1 |
| 4 | LED | | 1 |
| 5 | Connecting wire | | ADEQUATE |
| | | | ADEQUATE |

**Theory -** A BCD counter counts in binary coded decimal from 0000 to 1001 and back to 0000 Because after of the return to 0 after a count of 9, a BCD counter does not have a regular pattern as in a straight binary count.

Synchronous BCD counter can be cascaded to form a counter for decimal number of any length.

**Result -** The operation of BCD counter has been verified

**Precautions -** 1. Switch off the power supply while making connections.

Teacher's Signature : _____

Name of Experiment..... Date..1-12-22

Experiment No..9 Experiment Result..... Page No.24

# Experiment NO-9

Object - To study of 7 segment decoder.

Apparatus required- Bread board, IC-7447, LED connecting wires.

Theory- IC 7447 is a BCD to decimal decoder which takes BCD as input and convert.

It provides seven output pair which delivers data

Truth table for 7 segment.

| Decimal digit | a | b | c | d | e | f | g |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 2 | 1 | 1 | 0 | 1 | 1 | 0 | 1 |
| 3 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 4 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 5 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |
| 6 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 7 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 8 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 9 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |

Teacher's Signature :

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No

Precaution-

1. Connection should be proper and accurate
2. See the LED's output and note it.
3. Check power supply before applying.
4. Verify the truth table

Result- The operation of 7 segment decoder has been verified.

Teacher's Signature :